

Requirements

Change Log:

Date	Version	Sections Changed	Summary of Changes	Reasoning	Author(s)
2026-01-30	v1.1	Problem Statement and Scope	Changed language to reference two ML models instead of one	Architecture diagram and stakeholder presentation	Everyone
2026-01-30	v1.1	Consolidated Requirements	Removed the implication that large lists will produce a “learn more” dialogue from REQ-002	Fall term project partner meeting	Hailey and Grant
2026-01-30	v1.1	Consolidated Requirements	Changed the language of REQ-004 and REQ-005 to match the current UI	The tab names on the current implementation of the UI	Hailey
2026-01-30	v1.1	Ethics, Risks, and Constraints Traceability	Deleted the risk of IP licensing violation	Project partner priorities (stakeholder memo)	Everyone

Problem Statement & Scope

Problem Statement: Utilize the database of 40,000 species name-mass pairs supplied by Professor Novak's lab to produce two machine learning models which accurately predict the body mass of an input species from its scientific name and taxonomy. These models will be accessible via a web interface for single and multiple query requests.

Primary users: Biology and ecology researchers.

Secondary users: Programmers using data scraping tools.

Context of Use: The web interface / API will be used by predominantly researchers in the context of lab work. They may need to find the mass of one or more species, which is a vital measure for various fields in the science.

System Boundaries: Internal elements would include the webpage frontend, a backend for API calls, and the two mass prediction models (one an LLM-based implementation, and one a random forest-based implementation). External elements would include the catalog of life data, the body mass data that Professor Novak's lab compiled, and visualized phylogenetic tree data.

Success Criteria:

- 1) The model will pass verification of test data with at least 95% accuracy.
- 2) The web interface will handle at least 10 simultaneous queries.

Progress Toward Success Criteria:

- The model is operational, but does not yet produce results within 95% accuracy over the test data. We are currently working on several angles to improve the model such that it reaches this bar.
- As for the ability for the web interface to handle at least 10 simultaneous queries, this is dependent on the backend infrastructure first. As far as the model capabilities, it can make these predictions very quickly. However, we still need to fully connect the model to the frontend using our existing system architecture. After this connection is made, more direct progress can be made in this success criteria.

Non-Goal: Efficiency/Optimized response time

Consolidated Requirements

Functional Requirements:

REQ-001: The model will return a confidence interval for all predicted body masses. (high)

AC: Given a user has input a species name or list of species names, when the predicted body mass is calculated and displayed, then a corresponding confidence interval will also be displayed.

Status: Incomplete

REQ-002: The webpage will respond to a user's input by displaying an explanation of how the model came up with the body mass estimate(s) (i.e. was it predicted or directly sourced from database) (high)

AC: Given a user has input a species name or list of species names not exceeding 5 species, when the predicted body mass is calculated and displayed, then an explanation of how the prediction was reached will also be displayed.

Status: Incomplete

REQ-003: The user is able to input multiple search queries (10+) simultaneously (high)

AC: Given an input list of 10 species, when a user submits it to the web interface, then the model outputs the mass of said 10 species.

Status: Incomplete

REQ-004: The web interface will contain a help page to help novice users. (low)

Status: Partially Complete - This page exists, but the content is not finished

Commit: <https://github.com/marknovak/TaxonBodyMassML/commit/804a709>

REQ-005: The web interface will contain a "more information" tab to allow users to learn more about the algorithms and to provide appropriate citations. (low)

Status: Partially Complete - This page exists, but the content is not finished

Commit: <https://github.com/marknovak/TaxonBodyMassML/commit/16b52cb>

REQ-006: The user interface will meet WCAG accessibility requirements (med)

Status: Incomplete

Non-functional requirements:

REQ-007: Transparency: the data used for the model must be public, no private data shall be used. (med)

Status: Complete - We have used only public data to train our models

PR: <https://github.com/marknovak/TaxonBodyMassML/pull/1>

REQ-008: Accuracy: the model will have an accuracy of at least 95% of body masses in the same order of magnitude

AC: Given a set of 100 species that are not already in the database, when a user inputs them into the web interface, then at least 95 must be within the correct order of magnitude.

Status: Incomplete

REQ-009: Updatability: the architecture must allow changes to the model to account for the addition of new measured body mass data.

AC: Given a set of new body mass data, when the GitHub is updated with that data, then the workflow should retrain a new model with the new body mass data.

Status: Incomplete

REQ-010: Repeatability: the model will give consistent body mass outputs for a given input.

AC: Given a particular species or set of species, when a user requests its body mass, the output must be the same every time.

Status: **Complete** - The current implementation returns consistent results for the same input

Commit: [Decision Tree Regression v1](#)

Prioritization Method & Results

Requirement	MoSCoW Rating
REQ-001	Must Have (project partner mandated)
REQ-002	Must Have (project partner mandated)
REQ-003	Should Have (is only achievable after baseline implementation)
REQ-004	Could Have (not vital to the project, but good to aim for)
REQ-005	Should Have (important in scientific claims; after baseline implementation)
REQ-006	Could Have (not vital to the project, but good to aim for)
REQ-007	Must Have (project partner mandated)
REQ-008	Must Have (project partner mandated; may be impacted by repeatability)
REQ-009	Should Have (important in working with scientific data)
REQ-0010	Must Have (project partner mandated; may be impacted by accuracy)

Ethics, Risks, and Constraints Traceability

- Allowing users to make larger queries may limit accessibility to other users because of limited computational resources.

Impact: High Likelihood: Med

Mitigation Steps: Ensure that the deployment server is using robust hardware.

Affected Requirements: REQ-003

- Accuracy is dependent on the quality of the data. A potential risk is that, if the quantity of data is insufficient to produce an optimal solution which meets our accuracy requirements, then our model could not possibly be better.

Impact: High Likelihood: Low

Mitigation Steps: Determine accuracy data from test data extracted from the dataset.

Affected Requirements: REQ-008

- Depending on the model implementation, exact repeatability may not be achievable due to underlying stochastic constraints with predictive modeling.

Impact: Low Likelihood: High

Mitigation Steps: Accept variance within a range of grams.

Affected Requirements: REQ-010

- Depending on the model implementation, it may not be possible to automatically generate the confidence interval.

Impact: Med Likelihood: Low

Mitigation Steps: Create a separate analytical process which is able to produce a confidence interval by assessing the model's prediction behavior.

Affected Requirements: REQ-001

Appendix

Merge Methodology & Change References

We each pasted our top-10 lists into a document and discussed our priorities on a voice call. We narrowed down our options by:

- 1) Highlighting favorites
 - a) Each person looked at all three lists and highlighted the requirements that they wanted to be included in the merged list, this reduced the amount of items we had to comb through.
- 2) Prioritizing duplicates
 - a) If more than one person included it in their list, it was important enough to include in the merged list. We tried to merge our writing styles, but typically just voted on the "best-sounding" one.
- 3) Prioritizing our project partner's wishes
 - a) Some of these requirements(like the confidence interval, explanation, and public data use) were explicitly outlined as priorities by our project partner
- 4) Trying to find a good mix of NFRs
 - a) Our lists contained more functional requirements than not, so we kept most of the non-functional requirements to balance this out

Consolidated Requirements (with unique IDs)

For high-priority items, include acceptance criteria directly under the item.

Include a short NFRs subsection.

Functional Requirements:

REQ-001: The model will return a confidence interval for all predicted body masses. (high)

AC: Given a user has input a species name or list of species names, when the predicted body mass is calculated and displayed, then a corresponding confidence interval will also be displayed.

REQ-002: The webpage will respond to a user's input by displaying an explanation of how the model came up with the body mass estimate(s) (i.e. was it predicted or directly sourced from database) (high)

AC:

Given a user has input a species name or list of species names, when the predicted body mass is calculated and displayed, then an explanation of how the prediction was reached will also be displayed.

REQ-003: The user is able to input multiple search queries (10+) simultaneously (high)

AC: Given a user inputs more than 10 species, when a user inputs into web interface, then model outputs the mass if said 10 species.

REQ-004: The web interface will contain an information tab to help novice users. (low)

REQ-005:

The web interface will contain a citations and methods tab. (low)

REQ-006: The user interface will meet WCAG accessibility requirements (med)

Non-functional requirements:

REQ-007: Transparency: the data used for the model must be public, no private data shall be used

REQ-008:

REQ-009:

REQ-010: (med)

REQ-008: Accuracy: the model will have an accuracy of at least 95% of body masses in the same order of magnitude

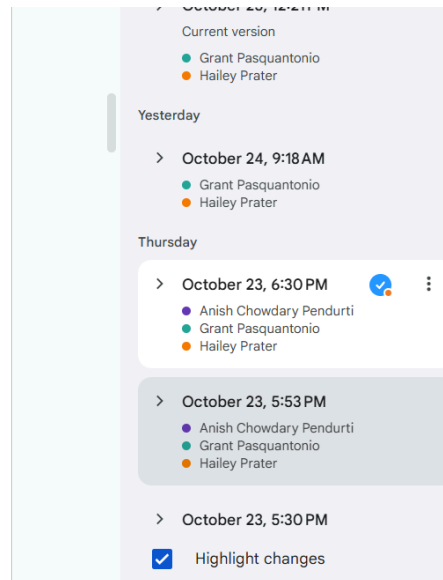
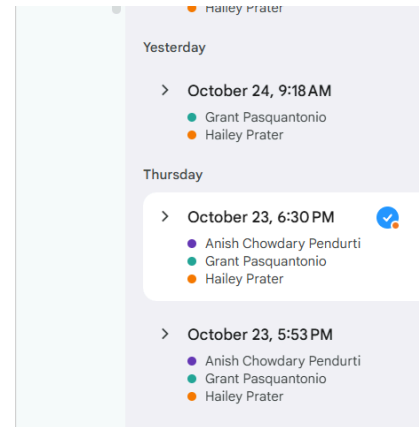
AC: Given a set of 100 species that are not already in the database, when a user inputs them into the web interface, then at least 95 must be within the correct order of magnitude.

REQ-009: Updatability: the architecture must allow changes to the model to account for the addition of new measured body mass data.

AC: Given a set of new body mass data, when the GitHub is updated with that data, then workflow should retrain a new model with the new body mass data.

REQ-010: Repeatability: the model will give consistent body mass outputs for a given input.

AC:



Individual Top 10 Requirements

(We highlighted the ones we wanted to include in the merged list)

Grant Pasquantonio's top 10 requirements: (priority level: low - high)

1. The web interface must comply with usability heuristics. (high)
 - a. User-friendliness allows people with a variety of technical expertise to utilize the product.
2. The model will return information regarding whether the body mass is predicted or measured. (med)
 - a. This is helpful in case the researcher needs/wants to know exactly where the data came from.
3. The model will return a confidence interval for all predicted body masses. (high)
 - a. This will help the user determine how trustworthy the prediction is.
4. The web interface will be able to accept at least 10 simultaneous requests. (med)
 - a. Users will likely need access to many species body mass data, so they should be able to query for multiple at once.

5. Accuracy: the model will have an accuracy of at least 95% of body masses in the same order of magnitude. (high)
 - a. The model must be decently accurate, otherwise it isn't helpful for scientific purposes.
6. Updatability: the architecture must allow changes to the model to account for the addition of new measured body mass data. (high)
 - a. The model must be flexible enough to accept new data, otherwise it will become quickly obsolete.
7. Repeatability: the model will give consistent body mass outputs for a given input. (high)
 - a. This indicates limited stochastic elements in the model, improving user confidence in accuracy.
8. The web interface will maintain at least a 97% uptime after deployment.(med)
 - a. Accessibility is a key part of the function of the product. It must be available for researchers to use for the majority of the time.
9. The web interface will contain an information tab to help novice users. (low)
 - a. Providing the option to read about how to use the software accommodates various levels of technical literacy.
10. The web interface will contain a citations and methods tab. (low)
 - a. This tab will allow researchers to see where the initial data came from, and how predictions are being made.

Anish Pendurti's top 10 requirements:

1. The model will be trained on $\geq 40\,000$ species name–mass pairs to learn phylogenetic patterns.
 - a. Make sure the model is trained well enough to give good outputs
2. The user is able to input multiple search queries(10+) and receive them accurately
 - a. A function required by project partner
3. The data used for the model must be public, no private data shall be used
 - a. A function required by project partner
4. The webpage must be usable by researchers with a minimum technical knowledge
 - a. Accessible to everyone
5. The system must be able to query at least a million species
 - a. Large enough data pool for the model to work with
6. The model must have a 95% accuracy rate
 - a. In order for users to get accurate outputs
7. The webpage will be public and accessible within a second

- a. Usability of the webpage
- 8. The model will give out a confidence interval
 - a. Allow the user to see an accurate estimate of the model output
- 9. The model will be built using python.
 - a. A function required by project partner
- 10. Documentation will be written on how to use webpage and the model.
 - a. Make sure users are able to use the webpage and model accurately

Hailey Prater's Top Ten Requirements (Ordered by priority high to low)

- 1) The webpage will respond to a user's input by displaying (a) predicted body mass(es)
 - a) This is the entire point of and purpose behind our program
- 2) The webpage will allow users to input a single species name
 - a) This is the first trigger to further functionality
- 3) The webpage will respond to a user's input by displaying a calculated level of certainty/confidence in the model's prediction
 - a) This was a functionality prioritized by our project partner
- 4) The webpage will respond to a user's input by displaying an explanation of how the model came up with the body mass estimate(s) (i.e. was it predicted or directly sourced from database)
 - a) This was a functionality prioritized by our project partner
- 5) The model will display deterministic body mass outputs (i.e. entering the same species name twice will result in the same body mass result both times)
 - a) This is what will give users confidence in our model; they will know it is not just throwing out a random guess
- 6) The webpage will be publicly accessible
 - a) We want ecologists and other scientists outside of Mark Novak's circle to have access to this program
- 7) The webpage will allow users to input a list of species names
 - a) This was another thing specified by our project partner from the beginning, but since it leans more "ease of use", it is lower on the priority list
- 8) The model will maintain at least a 95% accuracy rate
 - a) Reliability is important if we are claiming to make scientific predictions
- 9) The user interface will meet WCAG accessibility requirements
 - a) We want users to be able to interact with our program regardless of computer-literacy
- 10) The user interface will be designed to suit both laptops and mobile devices
 - a) We want our program to be accessible to as much of an audience as possible

